# POKÉMON
# RED, BLUE AND YELLOW
# CUSTOM MOTHERBOARD




PRODUCT

HTTPS://SHOP.GILTESA.COM/PRODUCT/POKEMON-RED-BLUE-AND-YELLOW-FOR-GAME-BOY-CUSTOM-MOTHERBOARD

**PLEASE READ THROUGH THESE INSTRUCTIONS ENTIRELY BEFORE ATTEMPTING TO INSTALL.**

**WARNING: IF YOU ARE NOT COMFORTABLE WITH SOLDERING, OR PERFORMING ANY STEP IN THIS GUIDE, DO NOT PERFORM THE INSTALL YOURSELF. FIND SOMEONE WHO IS COMFORTABLE TO DO IT FOR YOU.**

## INDEX

DESCRIPTION ..... 4
FEATURES ..... 4
INCLUDED ..... 4
RECOMMENDED / REQUIRED [NOT INCLUDED] ..... 5
NOTES ..... 5
BOARD DETAILS ..... 6
INSTALLATION STEPS ..... 7
PRE INSTALLATION STEPS ..... 8
1. DISASSEMBLY THE CARD SHELL ..... 8
2. DISASSEMBLY THE CIRCUIT COMPONENTS ..... 10
INSTALLATION STEPS ..... 12
1. ISOLATION OF THE CIRCUIT ..... 12
2. SANDING THE CENTERING TABS [OPTIONAL] ..... 13
3. SOLDERING TUTORIALS ..... 14
4. SOLDERING THE SOLDER JUMPERS ..... 14
5. SOLDERING SMALL COMPONENTS ..... 16
6. SOLDERING THE FOUR CHIPS ..... 18
7. SOLDERING THE BATTERY HOLDER ..... 21
8. CLEAN THE CIRCUIT ..... 22
9. DONE! ..... 23
FREQUENTLY ASKED QUESTIONS - FAQ ..... 24

# DESCRIPTION

This is a **full-color printed custom motherboard** for the games **Pokémon Red, Pokémon Blue and Pokémon Yellow** for Game Boy.

You can use it to customize your original cartridge, whose battery has long since died or whose original PCB has become damaged or corroded.

Additionally, as you can see in the photos, this new circuit will give your Pokémon cartridge an incredible new look. Pair it with a transparent Game Boy or Game Boy Color shell to show off the artwork.

# FEATURES

- New full-color printed motherboard for Pokémon Red, Blue and Yellow.
- Compatible with original Nintendo **MBC3** and **MBC5** donor cartridges.
- Replace the battery without soldering whenever it needs replacing.
- Compatible with GBFlash, allowing you to back up your save before the transplant and restore it afterwards.

# INCLUDED

- 1x Full-color printed custom motherboard.
- 2x 10K 0603 resistors (1 required).
- 2x 33pF 0603 capacitors (1 required).
- 5x 10nF 0603 capacitors (4 required).
- 1x Battery holder.
- 1x Custom vinyl sticker for the back with a Pokémon-style design.

# RECOMMENDED / REQUIRED [NOT INCLUDED]

- An original Pokémon Red, Blue or Yellow donor cartridge matching the selected version.
- Transparent game cartridge for Game Boy or Game Boy Color. (1)
- New CR2032 button battery.
- 3.8mm gamebit screwdriver.
- Sandpaper or metal files (2)
- Tweezers.
- Tin soldering iron.
- Tin.
- Flux.
- Isopropyl alcohol.

(1) Remember to purchase the shell for the circuit. It's up to your discretion whether to use the one for Game Boy or Game Boy Color; both are compatible.

(2) Could be optional. In case it's necessary, it serves to reduce the side tabs that the circuit has. This depends on how faithful the shell is to the original; the tabs prevent the circuit from wobbling or shifting from side to side. If the circuit doesn't fit into the shell, you'll need to file the tabs slightly.

# NOTES

This motherboard is designed for the MBC3, MBC3A, MBC3B and MBC5 cartridge revisions used in most international releases. It is not compatible with the early Japanese MBC1 revisions. Please check the installation guide for full compatibility information before purchasing.

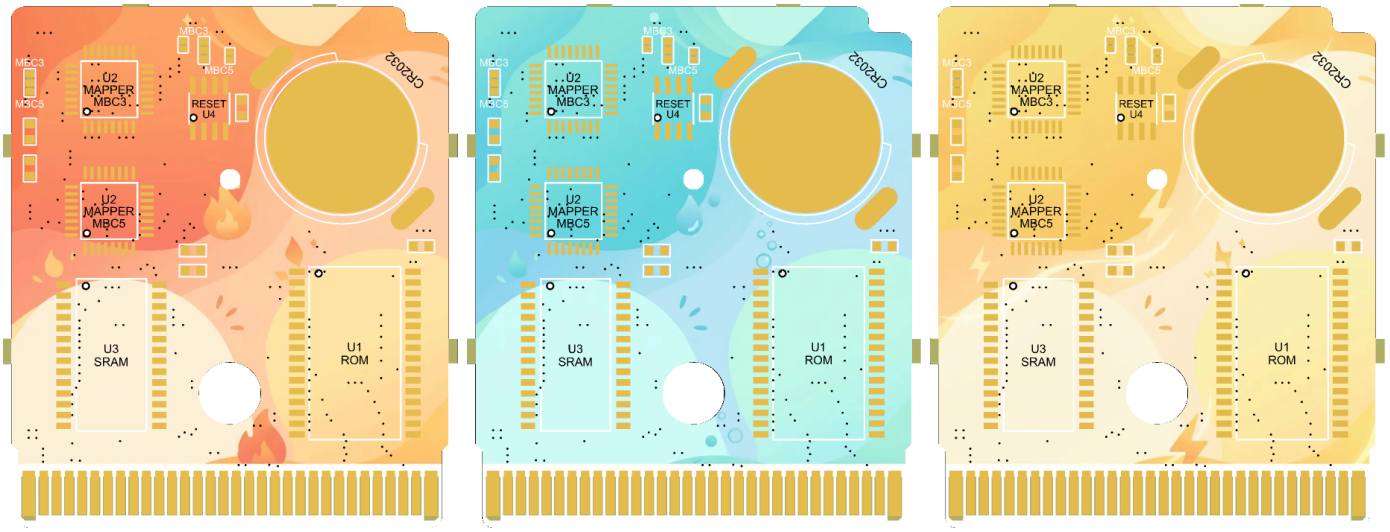
Although it is designed for use with Pokémon, it is also compatible with any other game whose motherboard has the identification number DMG-A02-01 or DMG-KFDN-10 and does not use an RTC.

# BOARD DETAILS

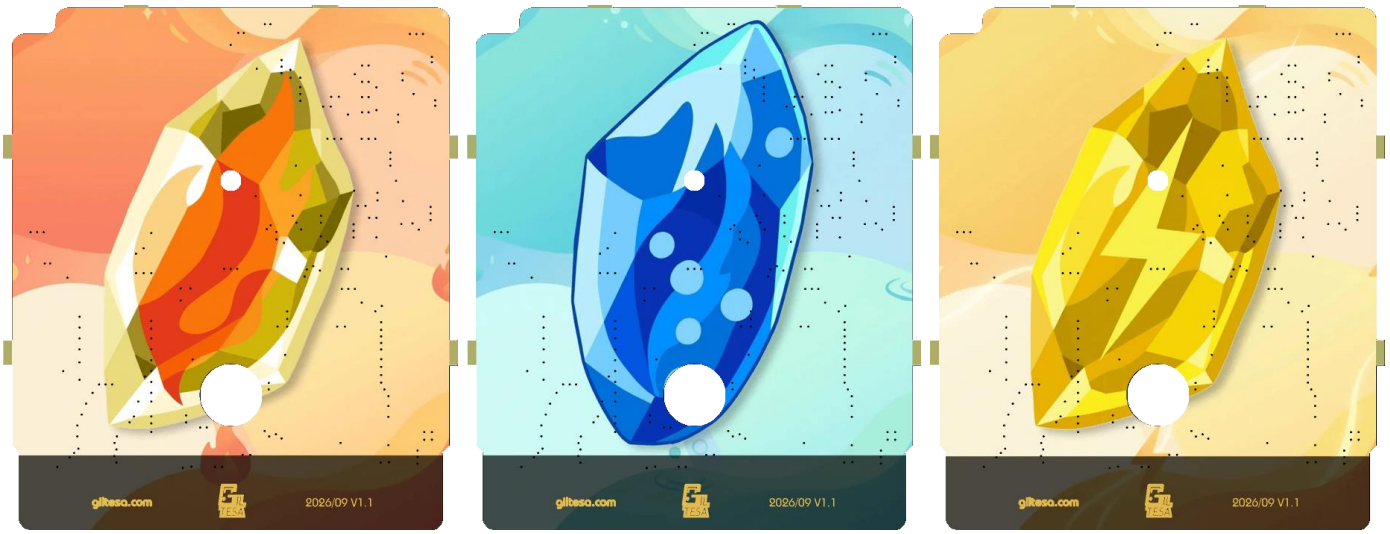
The board doesn't have any connectors or pads where wires need to be soldered like our other kits. However, this board doesn't come assembled; you will be the one to assemble the components supplied with the kit, as well as the 4 chips that make up the game, which you'll need to extract from your donor game.

All the steps are identical, regardless of the board color. In this guide, we will show you how to install it on Pokémon Blue.

*Light-colored front design that makes it easier to solder the components onto the board and see them clearly once the cartridge is assembled.*



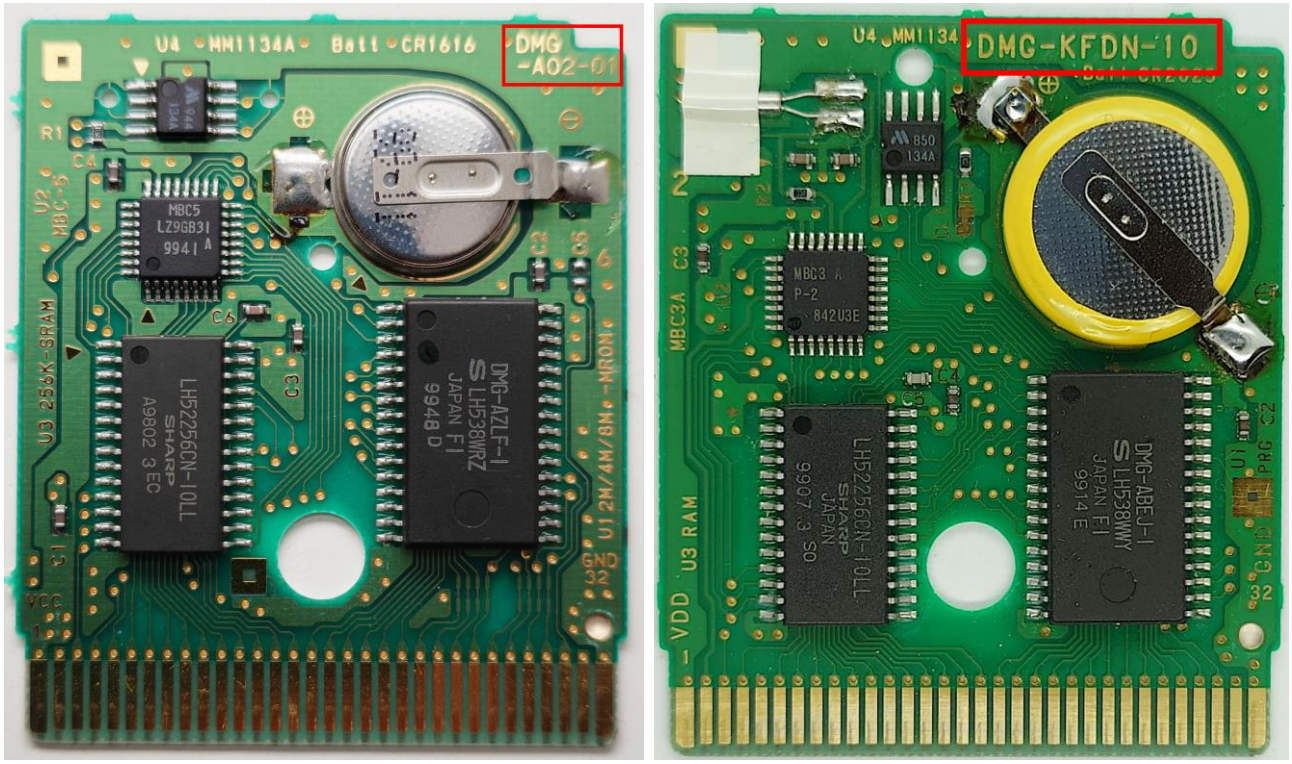

*Fire, Water, and Electric Evolution Stone design on the back. A sticker is included that can optionally be used to give it a more recognizable look associated with the franchise.*



# INSTALLATION STEPS

Please, carefully read the following steps for a successful installation.

You can use any Pokémon donor with an **MBC3** or **MBC5** chip from American or European games, respectively. You can check whether yours is compatible by looking at the motherboard number. It should be either **DMG-A02-01** or **DMG-KFDN-10**. Earlier Japanese Pokémon games that came with an MBC1 are not compatible.




*Circuit images from [https://gbhwdb.gekkio.fi](https://gbhwdb.gekkio.fi)*

**NOTE**: *The board should also be compatible with any other Game Boy game that uses a **DMG-A02-01** or **DMG-KFDN-10** and does not use the RTC/clock function.*

*So, even if the board design has nothing to do with the original game, it will allow you to restore it while continuing to use the game's original shell instead of a transparent one.*

# PRE INSTALLATION STEPS

Before proceeding with the installation, **you need to retrieve the 4 chips that make up Pokémon game from your game circuit**. These chips must be transplanted to the new circuit.

**This is where all the difficulty of the installation lies**, as it is most convenient to use a hot air soldering station or a heating table to extract the chips. It is not recommended to use just a soldering iron, although perhaps in combination with a desoldering pump, it may be possible. The choice of tools is up to you.

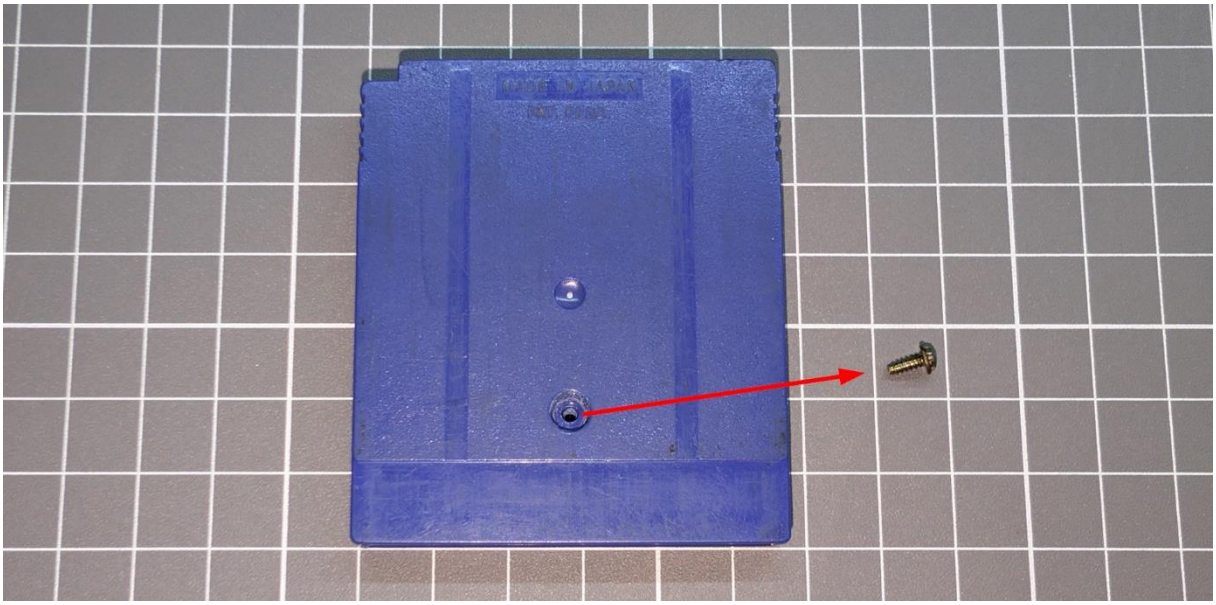
## 1. DISASSEMBLY THE CARD SHELL

First, you'll need to open the original game cartridge.

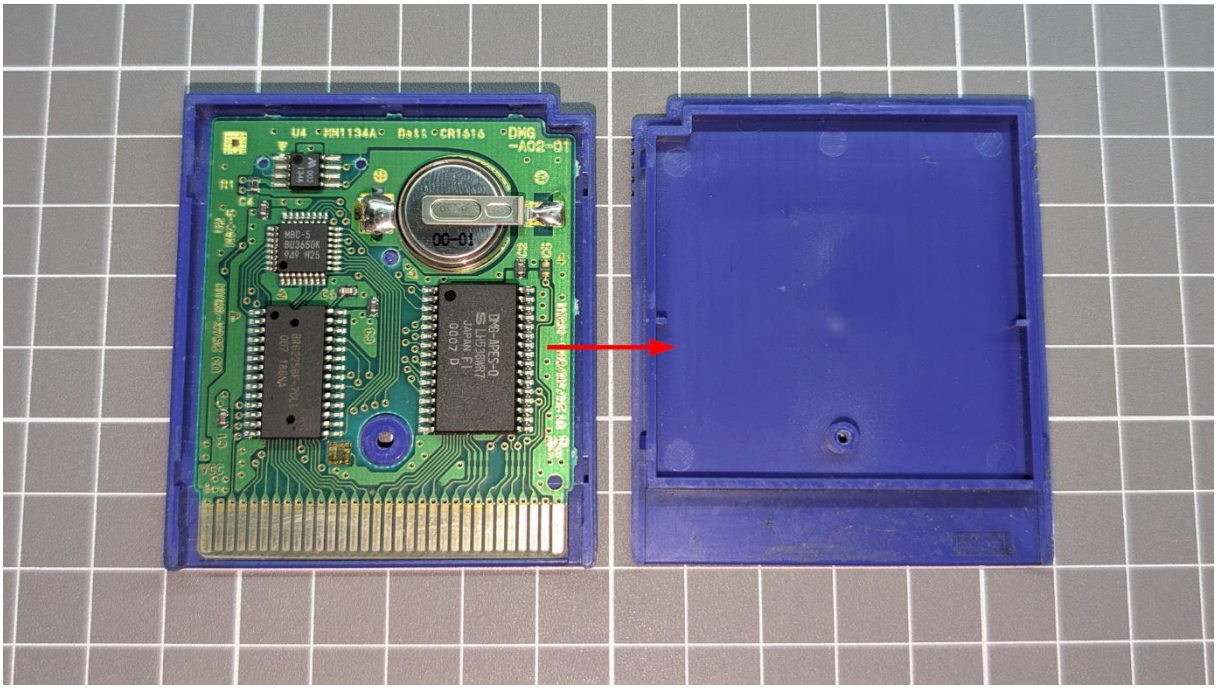



*Gamebit screwdriver*

The cartridge uses a **3.8mm gamebit screw**, this is the screw:



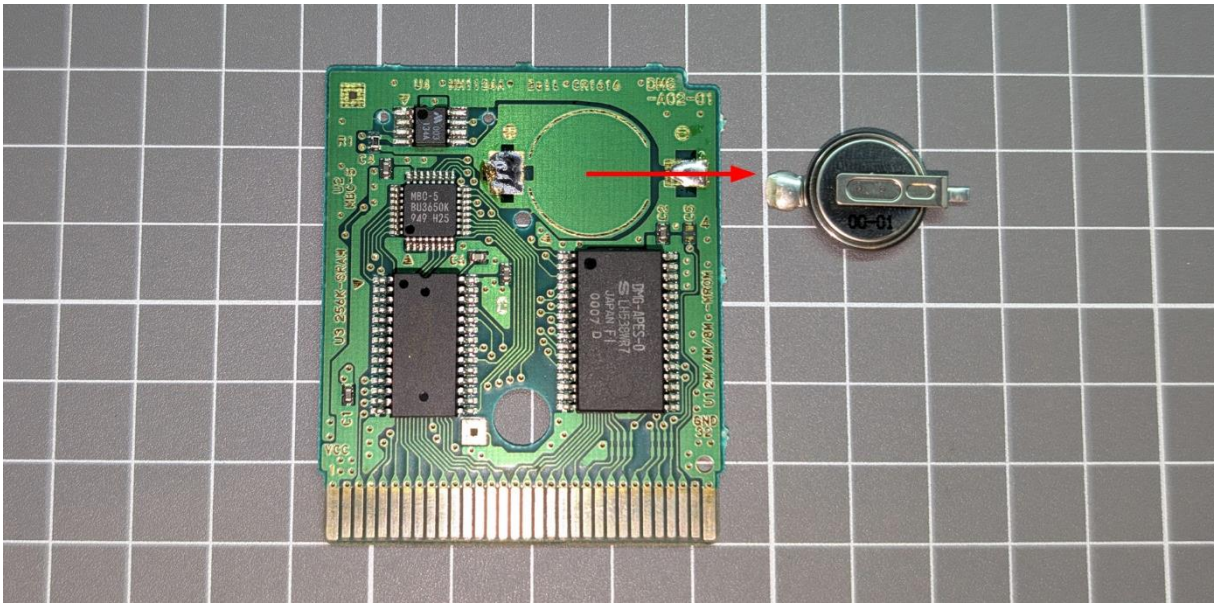
Once opened, you'll see the interior, just an electrical circuit where all the components are soldered.



## 2. DISASSEMBLY THE CIRCUIT COMPONENTS

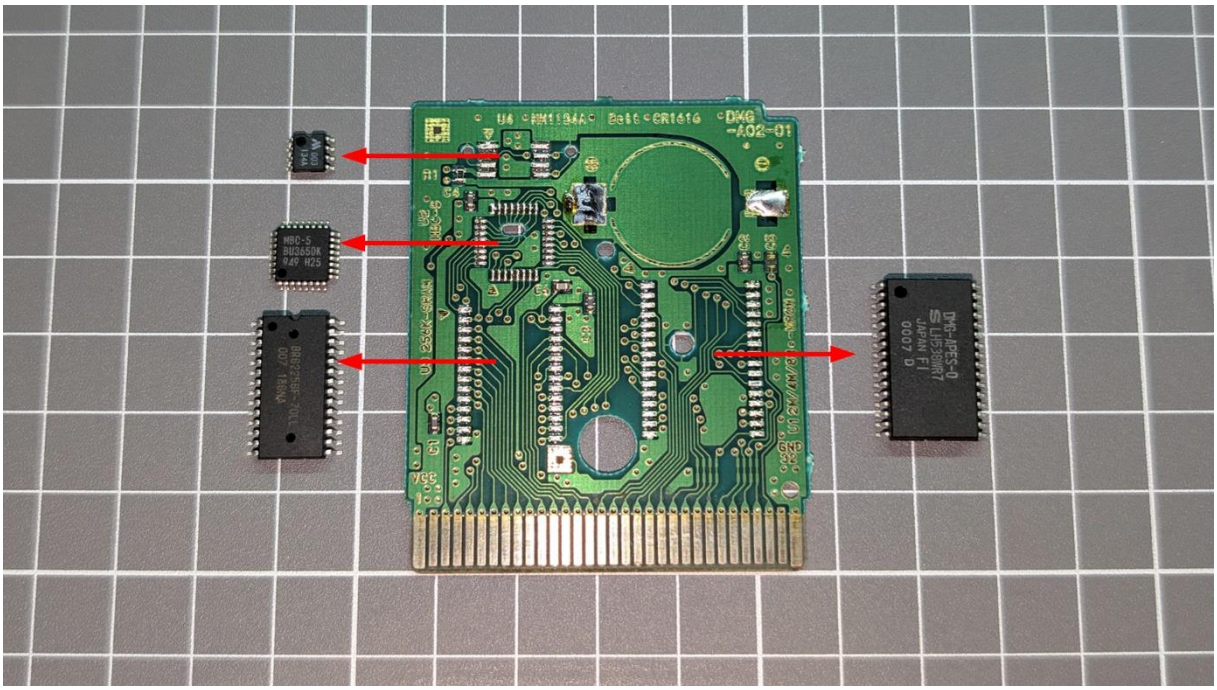
The first component to **remove is the game's battery**. No matter what type of tools you'll use to remove the rest of the components, **this component must be removed manually with a soldering iron.**

Optionally, you can use a heating plate. However, **if you use a hot air gun, do not apply hot air directly to the battery for safety reasons**. Instead, you can heat the board from underneath until it is warm enough to remove the battery with tweezers.



**IMPORTANT NOTE**: Removing the battery will erase any saved game data (if it hasn't already happened due to the battery running out). It is recommended to make a backup first if you don't want to lose your game progress.

Then comes the tricky part. It is recommended to use a hot plate or a hot air soldering station, in both cases heating the circuit from the underside so that the chips do not receive excessive heat. You can add a bit of flux to help the solder melt more quickly and thus free the chip to be removed with tweezers.



You don't need to remove the resistor or capacitors since new units are included in the kit.

# INSTALLATION STEPS

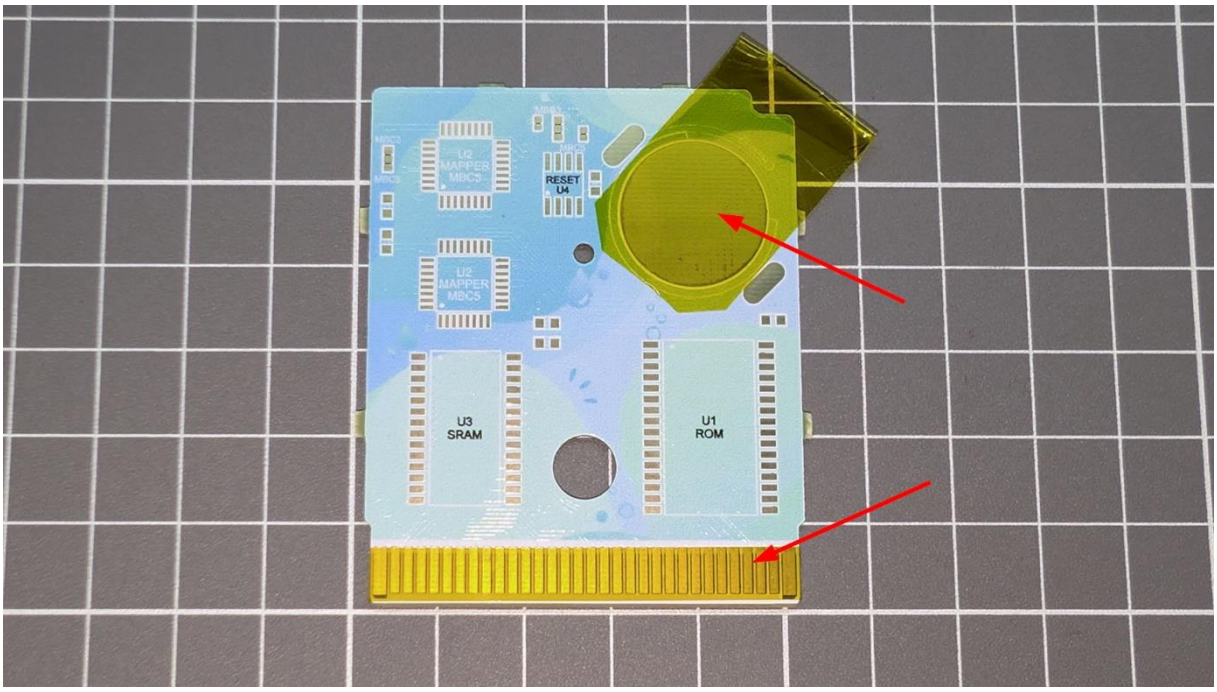
## 1. ISOLATION OF THE CIRCUIT

As you can see when opening the kit bag, the circuit already includes a kapton tape that protects the bottom where the pads are located, which will make electrical connection with the Game Boy game connector.

The negative terminal of the battery is also protected. Do not remove this until after soldering the battery holder.

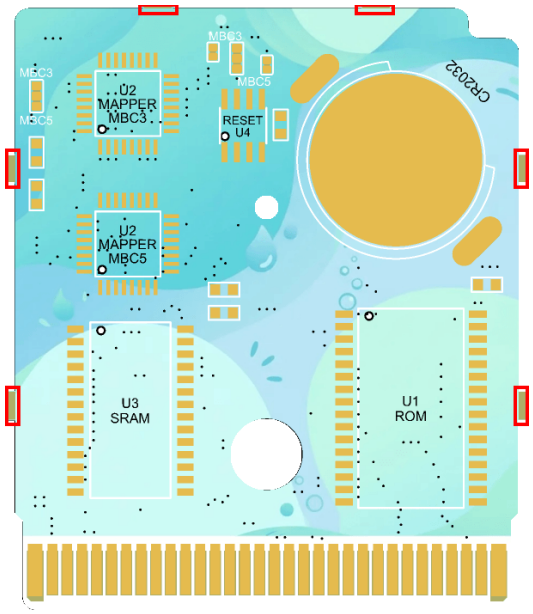
It's crucial to keep this kapton tape in place until you finish the installation completely. This prevents accidental solder from falling onto these pads and ruining the PCB.

If you have removed the kapton tape, put it back in place 😊



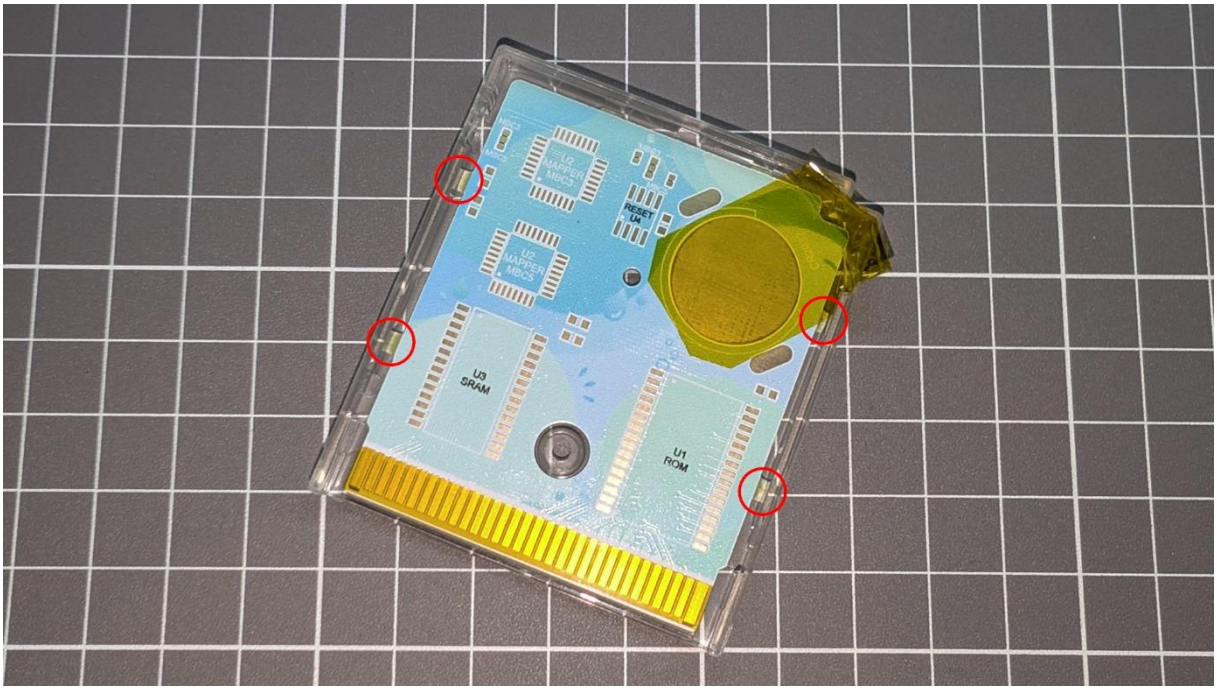
## 2. SANDING THE CENTERING TABS [OPTIONAL]

Since there's a wide variety of manufacturers producing card shells for Game Boy, not all of them have the same dimensions as the original shells. That's why the included circuit has side and top tabs to try to centre the circuit as much as possible within the shell.




**Insert the circuit into the shell you want to use and check if it fits correctly**. If you have trouble inserting it, you'll need to sand the tabs very slightly, first on one side and then on the other. Sand gradually and check if the circuit fits into place or not.

When you finish, clean the circuit with isopropyl alcohol.



## 3. SOLDERING TUTORIALS

We especially like these two videos, although there are countless of them on the internet:

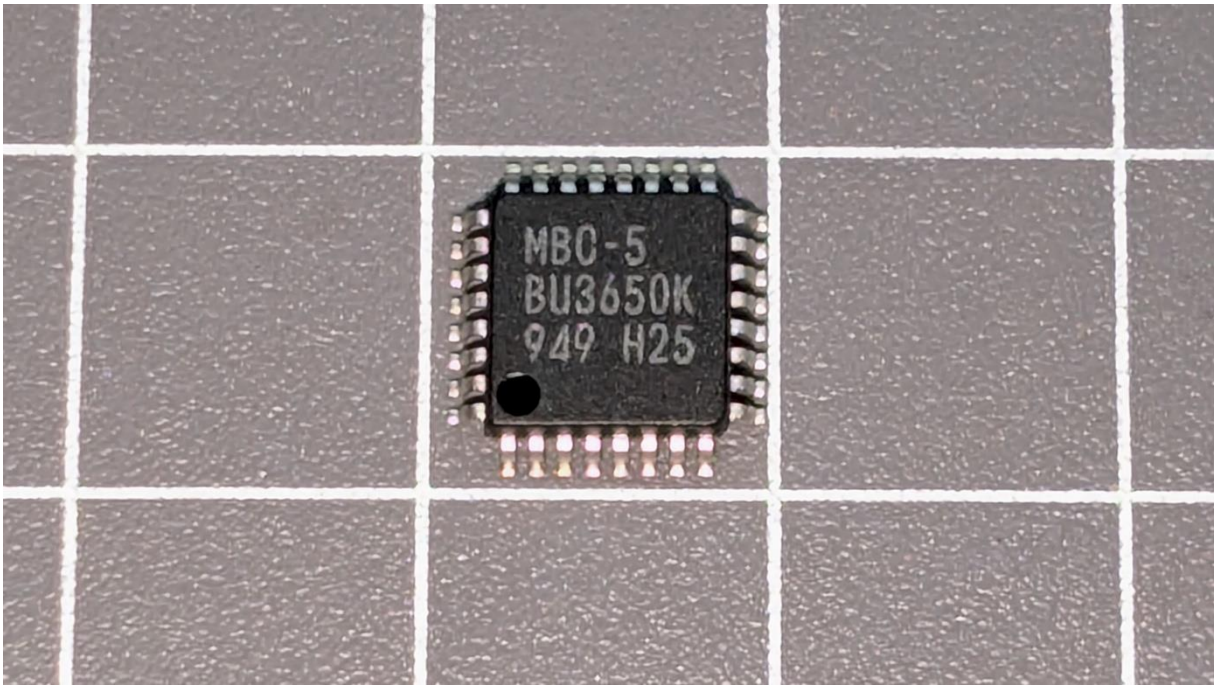
- SMD Soldering
- Professional SMT Soldering

## 4. SOLDERING THE SOLDER JUMPERS

The circuit is compatible with both the **MBC3** and **MBC5** mapper chips. However, some solder jumpers must be set to specific positions for the rest of the circuit to work correctly, depending on the MBC version of the donor game.

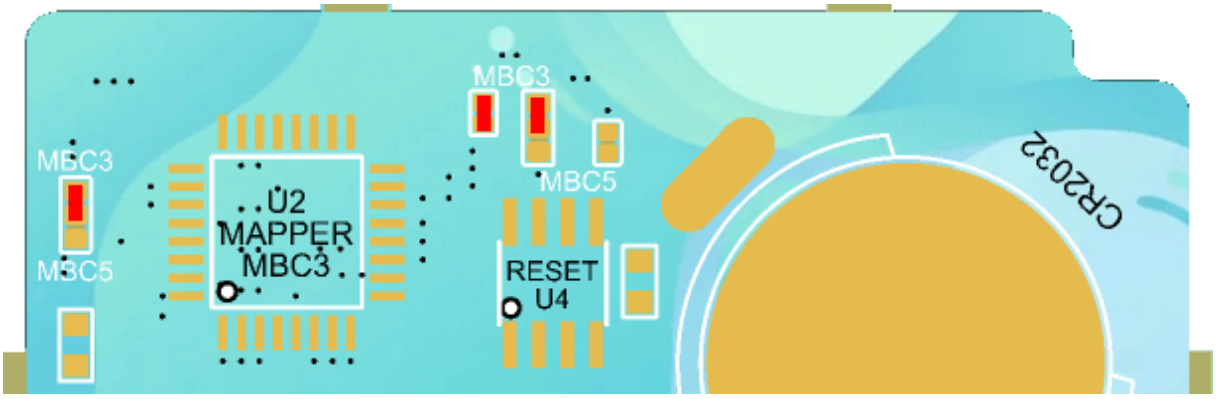
First, identify which MBC version you have. You can easily do this by reading the text on the top of the chip. You will see something like MBC3 (or MBC3A, MBC3B) or MBC5.

For our board, the mapper is an MBC5.

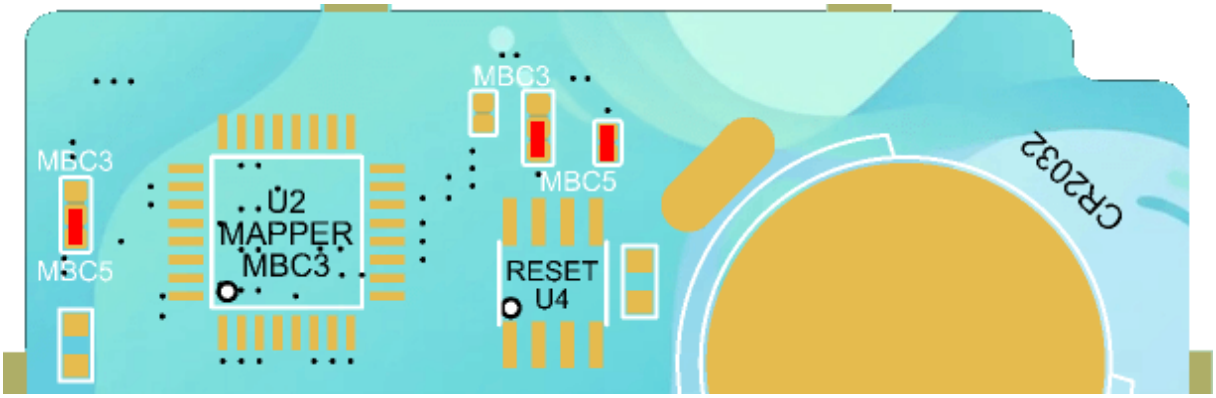


Based on this, solder the corresponding solder jumpers as shown in the images.

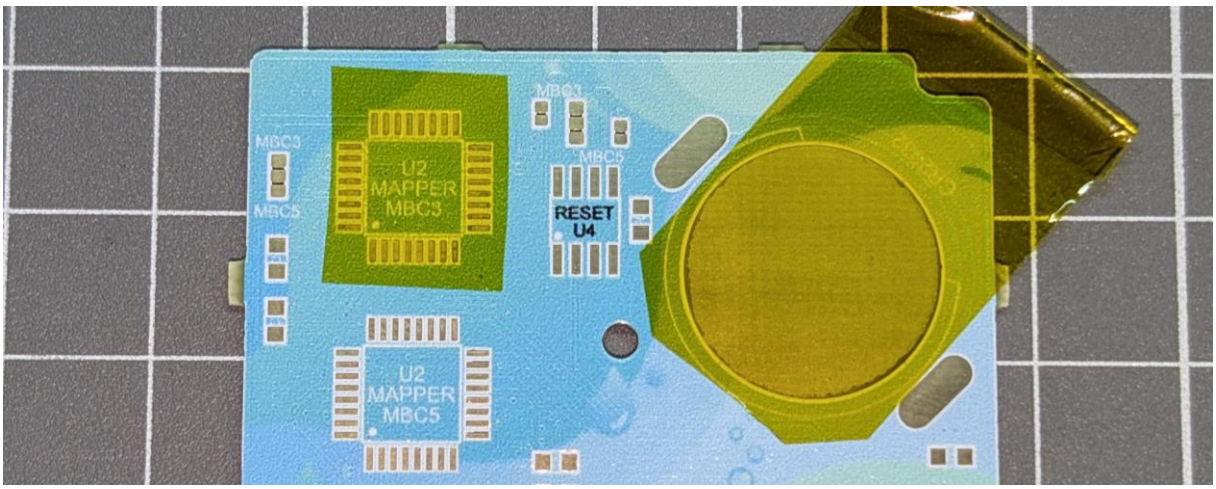
For the **MBC3, solder the solder jumpers in the upper position**, as shown in the image in red:




For the **MBC5, solder the solder jumpers in the lower position**.




NOTE: It is recommended to protect the MBC footprint where nothing will be soldered with kapton tape. This will prevent the pads from getting covered with solder and ensure a clean installation.



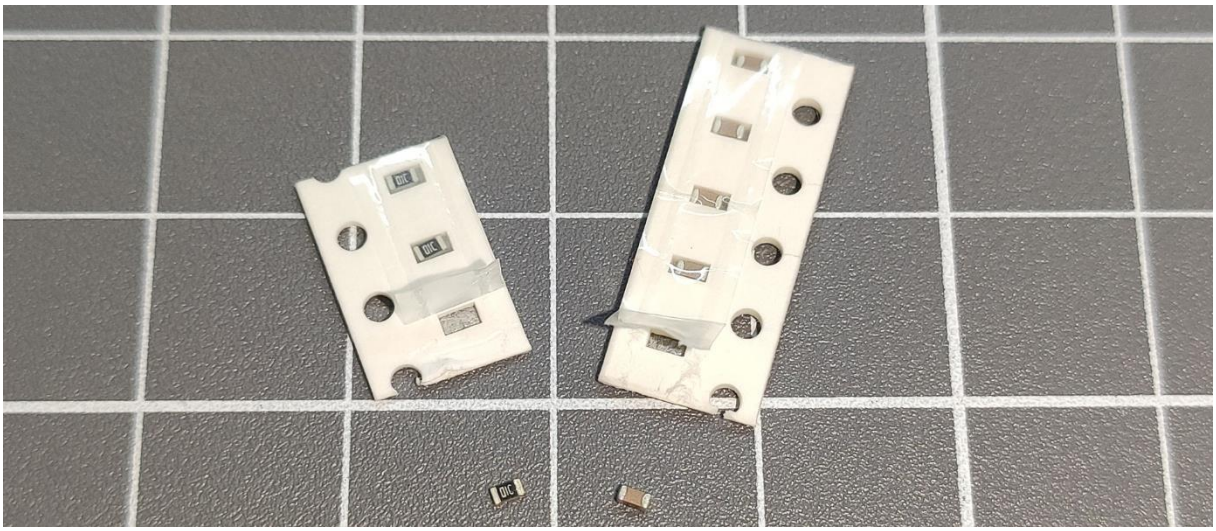

*In our case, we will be soldering an MBC5, so we cover the MBC3 footprint.*

## 5. SOLDERING SMALL COMPONENTS

It's time to proceed with the installation of everything. Start with the smallest components first, then the 4 chips in a specific order, and finally the battery holder.

The kit includes more resistors and capacitors than necessary because some might get lost during installation. Make sure to remove the components from the protective film one by one as needed and be careful not to lose them.

The **resistors are black** and have numbers written on them. The capacitors are brown.

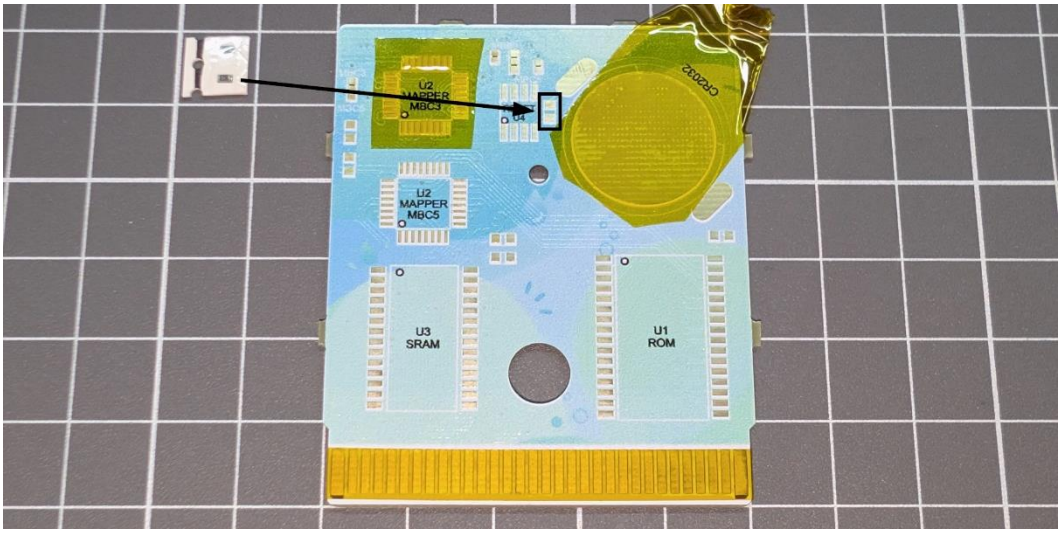


The circuit requires:

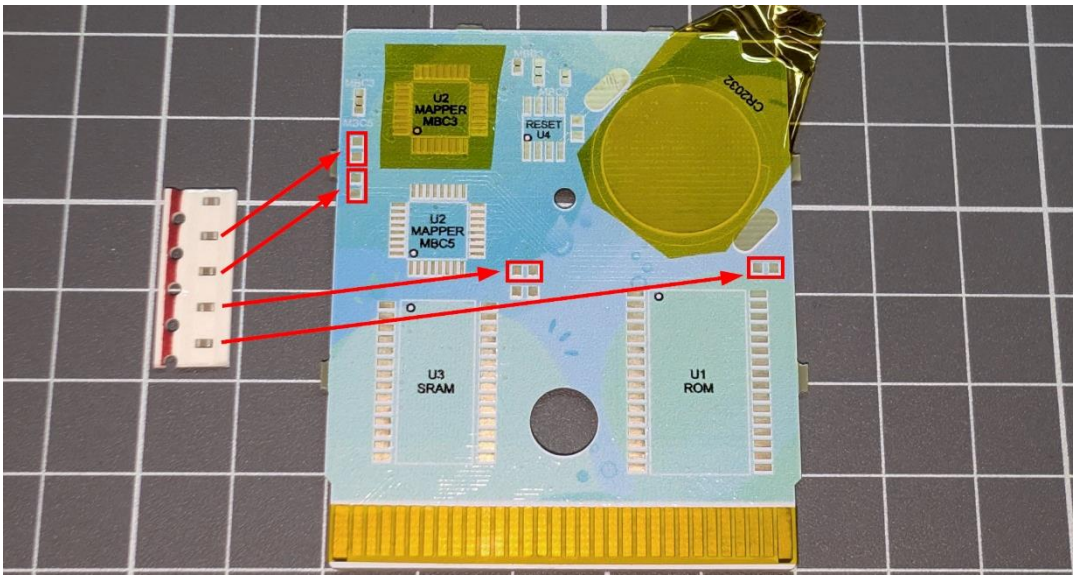
- 1x 10K resistor sized 0603.
- 4x 10nF capacitors sized 0603.
- 1x 33pF capacitor sized 0603.

The tape/cardboard packaging of each component is marked with a permanent marker to help you easily identify each component.

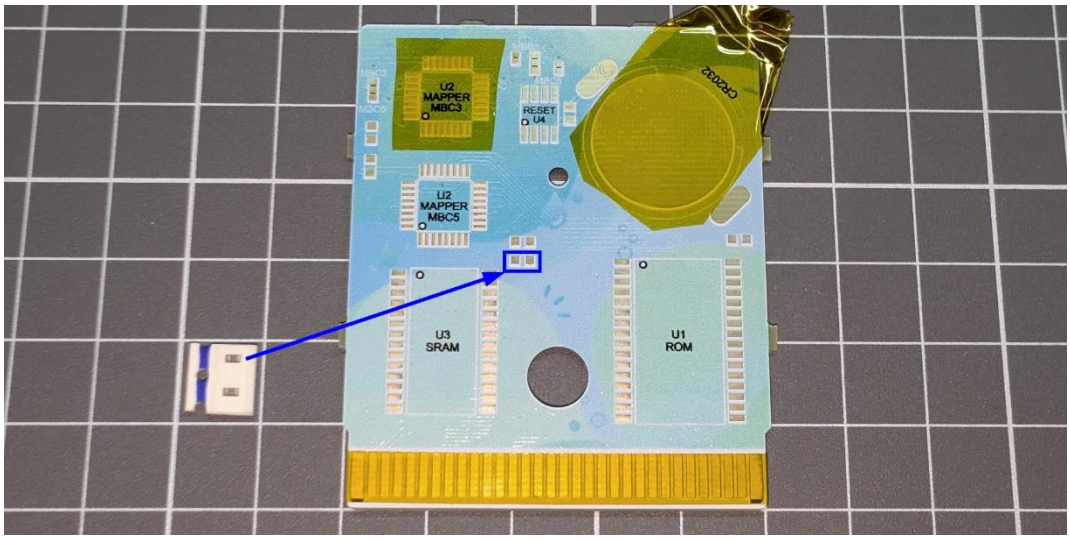
This is where you should solder the 10K resistor.




Here you should solder the 10nF capacitors.



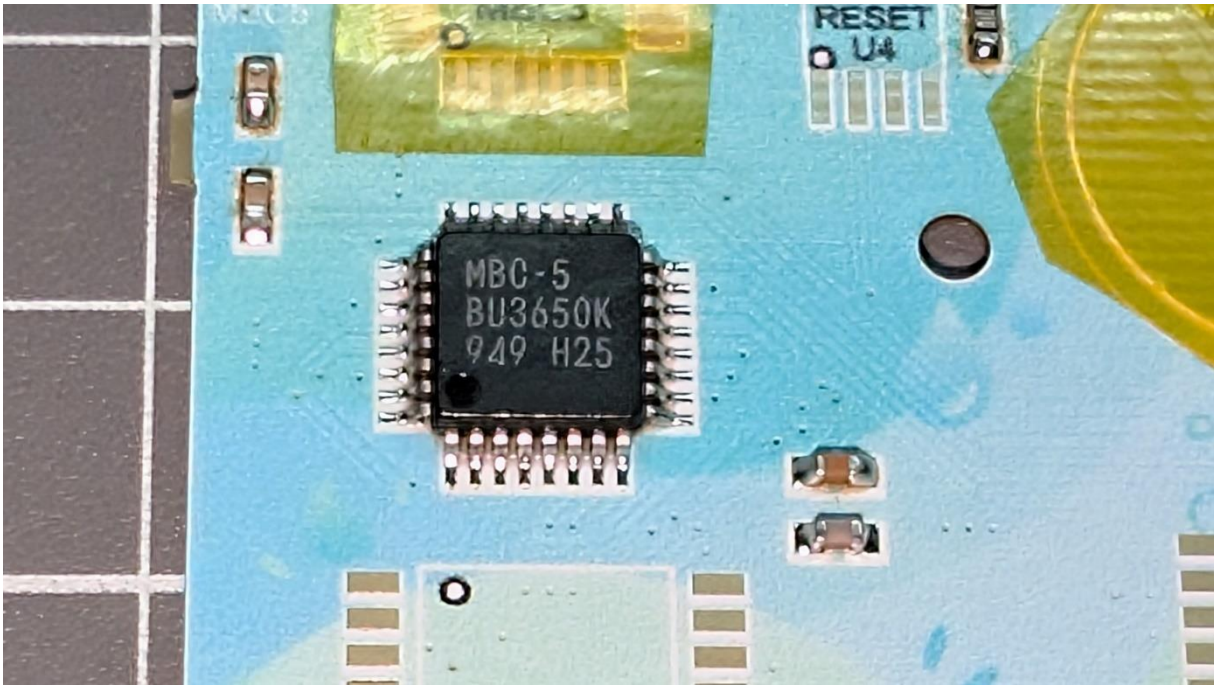

Finally, solder the 33pF capacitor here.



## 6. SOLDERING THE FOUR CHIPS

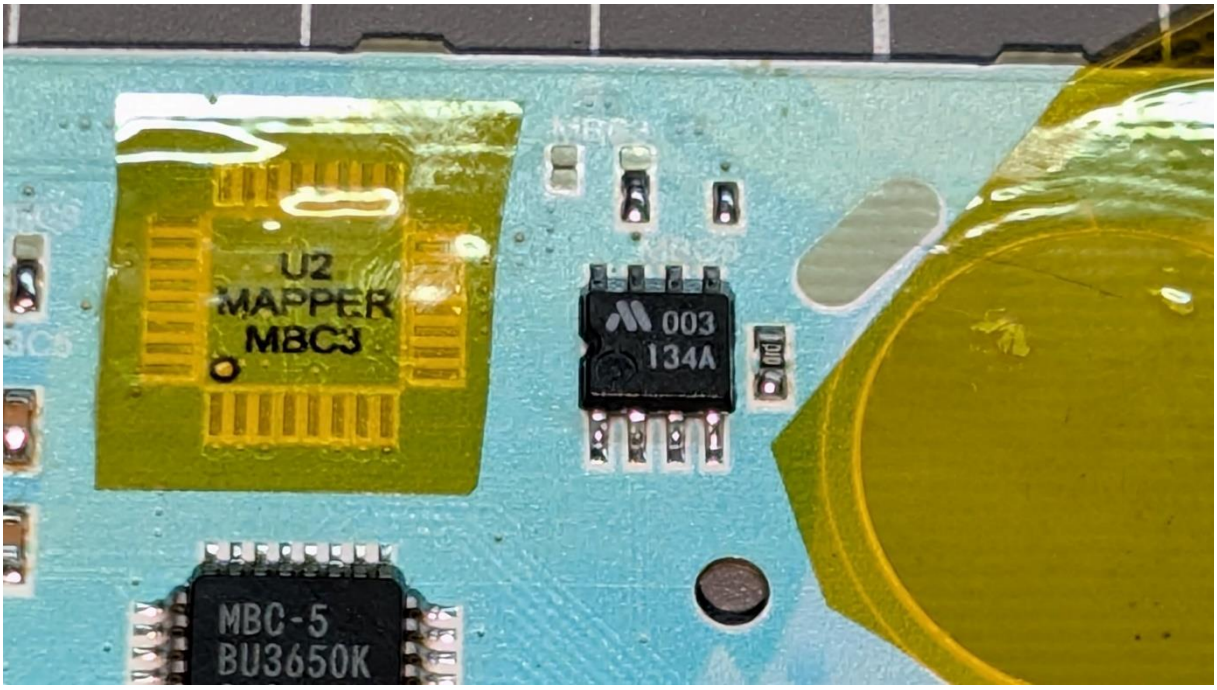
Start by soldering the four chips. In principle, the order doesn't matter, as they don't interfere with each other and there is enough space between them for soldering. However, it might be easier to start with the U2 MAPPER, as it has pins on all four sides.

IMPORTANT NOTE: When soldering each chip, make sure you do it in the correct position. You'll see that both the chip and the soldering site are marked with a circle indicating pin number one.

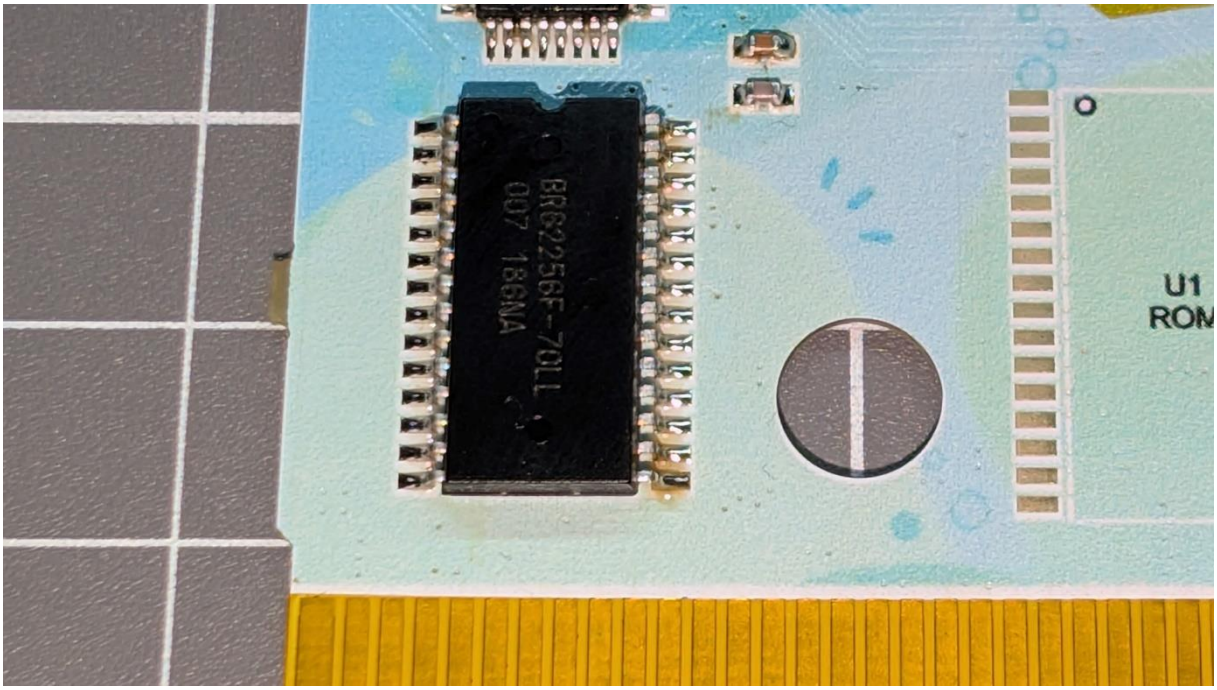



*U2 MAPPER*

Continue soldering the remaining chips in whichever order you prefer.



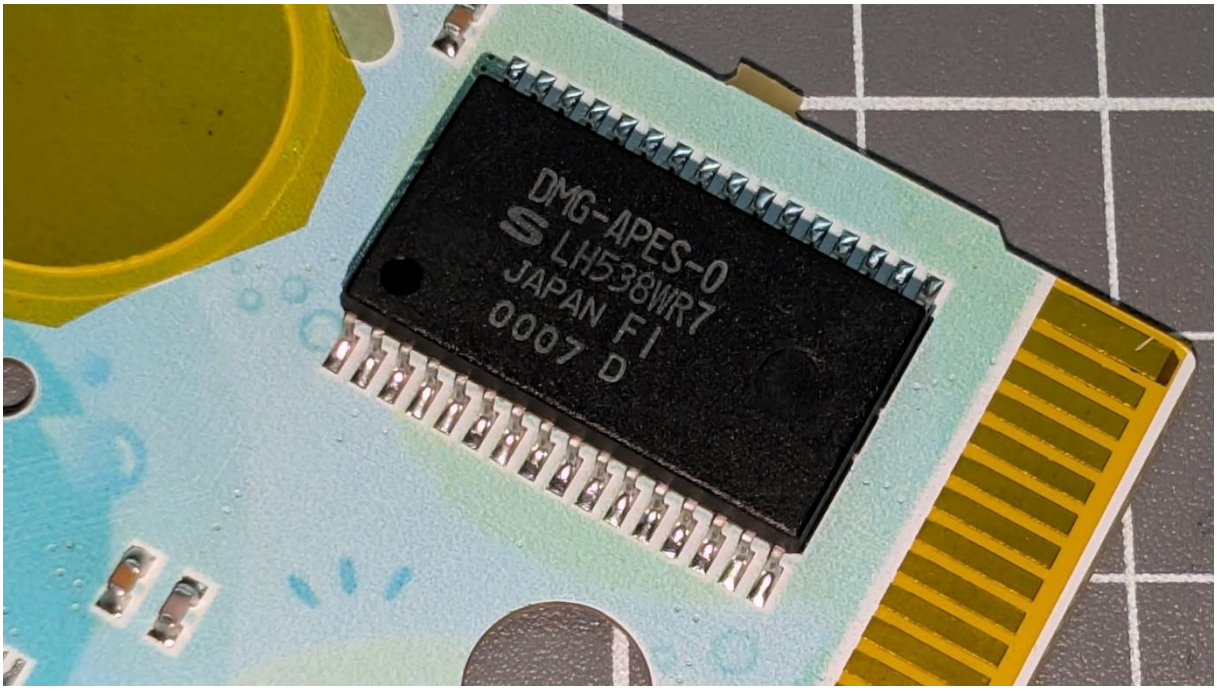

*U4 SRAM-SUPERVISOR*




*U3 SRAM*

Regarding the ROM chip, there are two versions: the European version, which has a larger chip with pins on the left and right sides, and the Japanese/American version, which has a smaller chip with pins on the top and bottom.

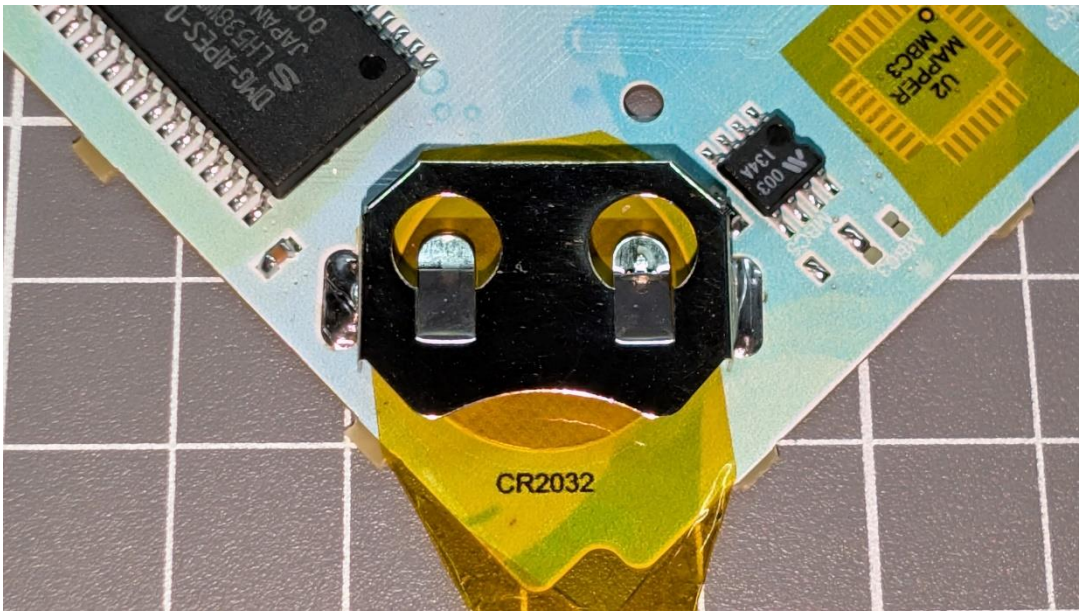
The PCB is compatible with both chip sizes. Just cover the pads you are not going to use with Kapton tape to prevent solder from getting onto them.



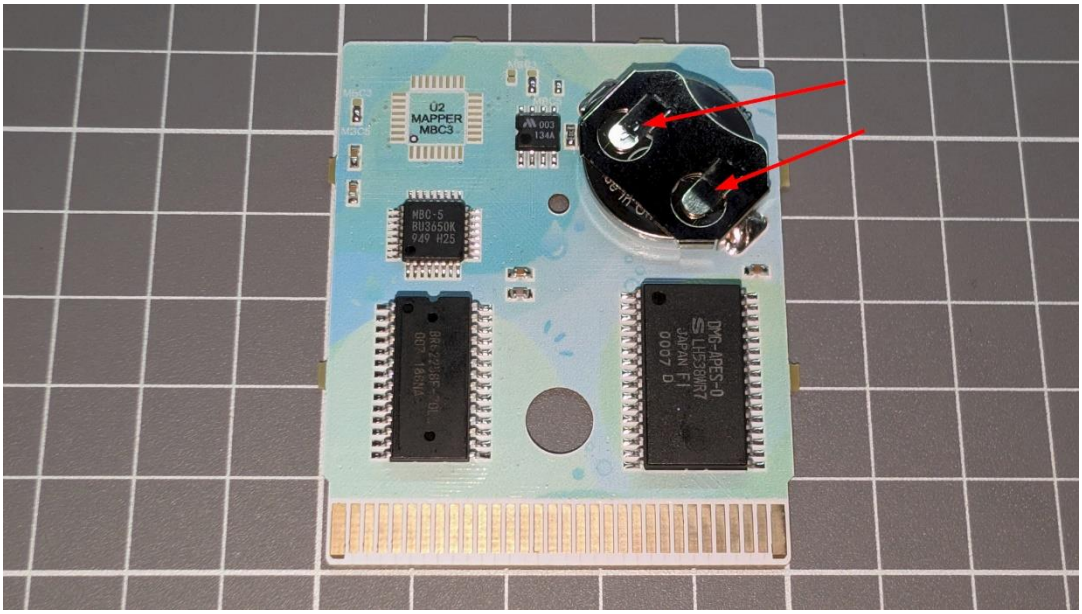

*U1 ROM*

## 7. SOLDERING THE BATTERY HOLDER

Solder the metal piece that will hold the battery in place. You must centre it properly, otherwise, the cartridge shell may not close afterward. If you prefer, you can solder one side first and check that it closes properly, then solder the other side to complete the installation.

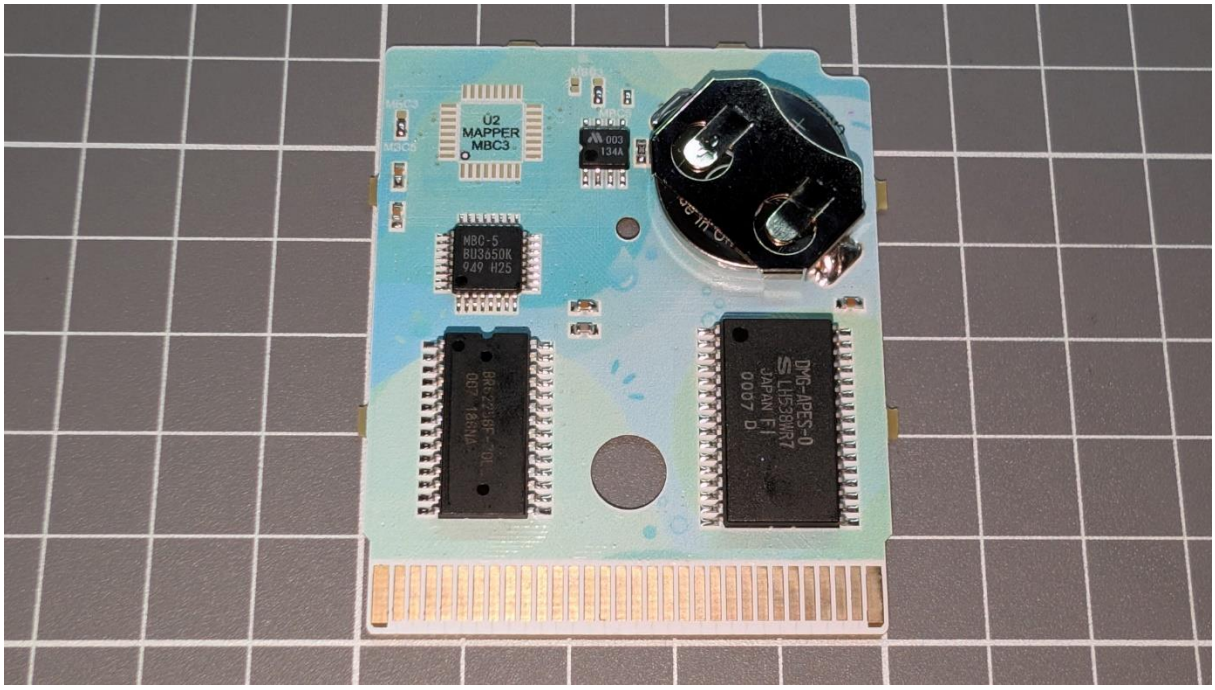



Once you have soldered both sides of the battery holder. Insert the battery into the battery holder. When you do this, you'll notice that the top tabs protrude too much. This prevents the cartridge shell from closing and may scratch it! You'll need to press them down slightly to make them flatter, you can use a flathead screwdriver to apply force and flatten the tabs.



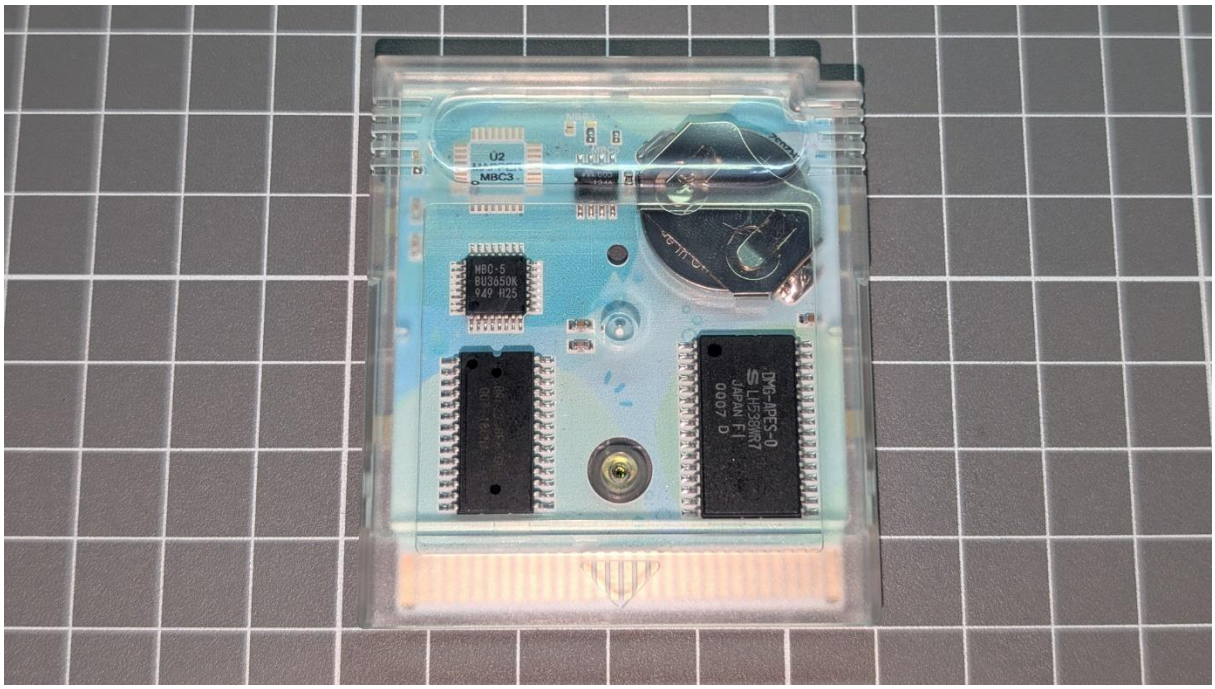
## 8. CLEAN THE CIRCUIT

The installation is complete. If you haven't done so during the installation of each component, it's time to clean the circuit with isopropyl alcohol.



## 9. DONE!

We now have our Pokémon game with its new colorful circuit, new battery, and transparent shell, making it look truly amazing!






# FREQUENTLY ASKED QUESTIONS - FAQ

1. ***THE GAME DOESN'T WORK OR DOESN'T FUNCTION CORRECTLY.***

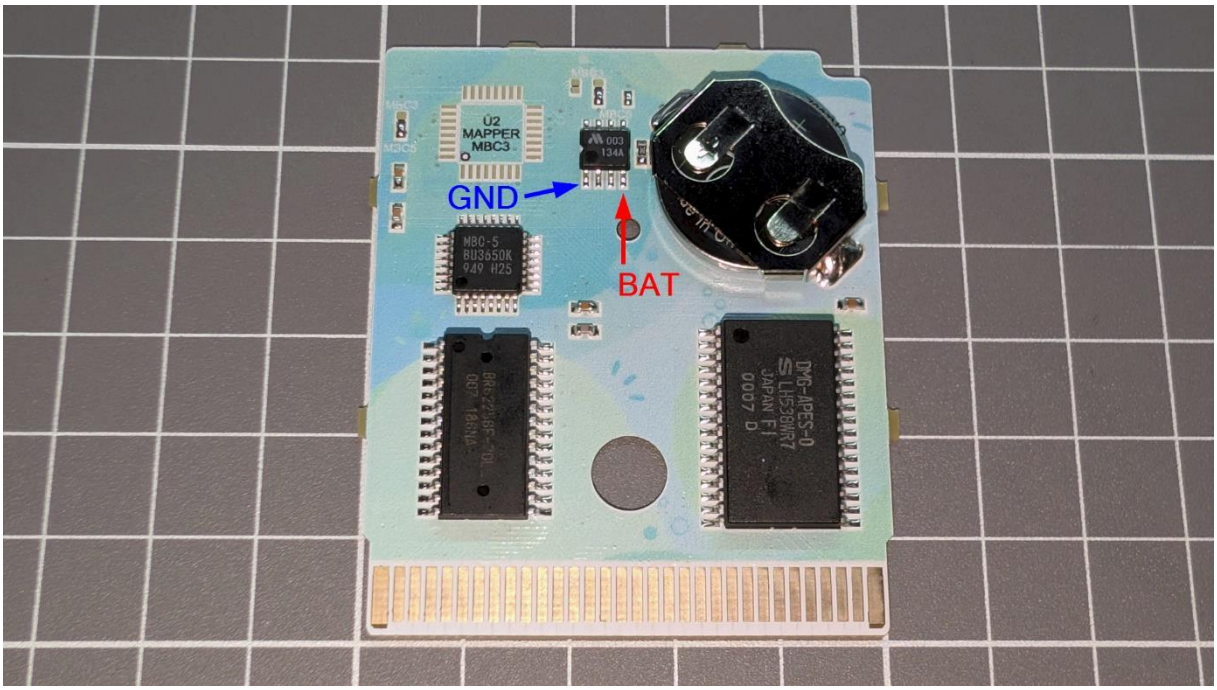
   Ensure that all chip connections to the circuit are properly made, that there are no cold solder joints, and that there are no short circuits between pins, etc.

   Use the following diagram to help you verify that each connection is correctly made. (*It's on the last page of the guide*)

2. ***THE GAME DOESN'T SAVE PROGRESS, AND WHEN YOU TURN ON THE CONSOLE, THE SAVED GAME IS LOST.***

   Ensure that the battery is not depleted, is inserted correctly (positive side up), and that the current is properly reaching the circuit.

   You can measure the voltage at these points with your multimeter to verify that the current is reaching the circuit:



### *3. WHAT IS THE INCLUDED STICKER FOR?*

The included sticker can be applied to the back of the PCB if you don't like the current design. This artwork was originally intended to be printed directly on the PCB, but manufacturing limitations made that impossible. For this reason, it is included as a sticker, and it's entirely up to you whether to apply it or not.

How to apply the sticker:

1. First, make sure the PCB is fully assembled and clean its surface with alcohol.

2. Use a black permanent marker to color all the outer and inner edges of the PCB, as the sticker is slightly smaller than the board.

3. Carefully peel back about 5 mm of the sticker's backing paper, then cut off the exposed section of backing paper with scissors so it doesn't get in the way.

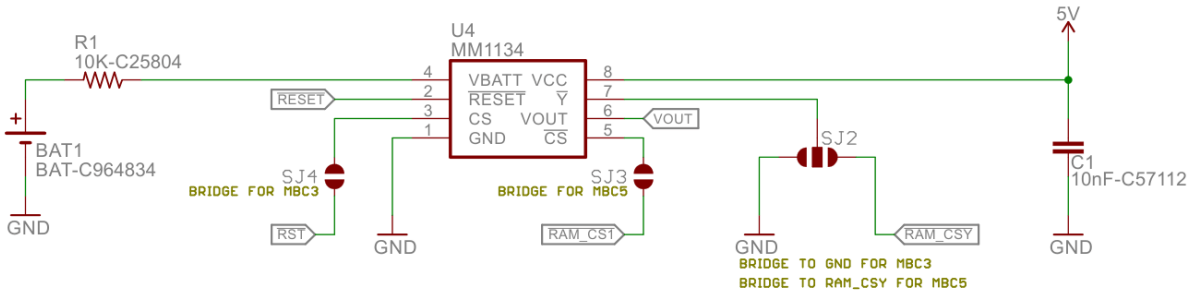
4. Align the sticker with the PCB, making sure it is properly centered. Gently press down on the exposed adhesive section to secure it in place.

5. Slowly remove the remaining backing paper while pressing the sticker down, working from the already attached side towards the opposite edge to avoid trapping air bubbles.

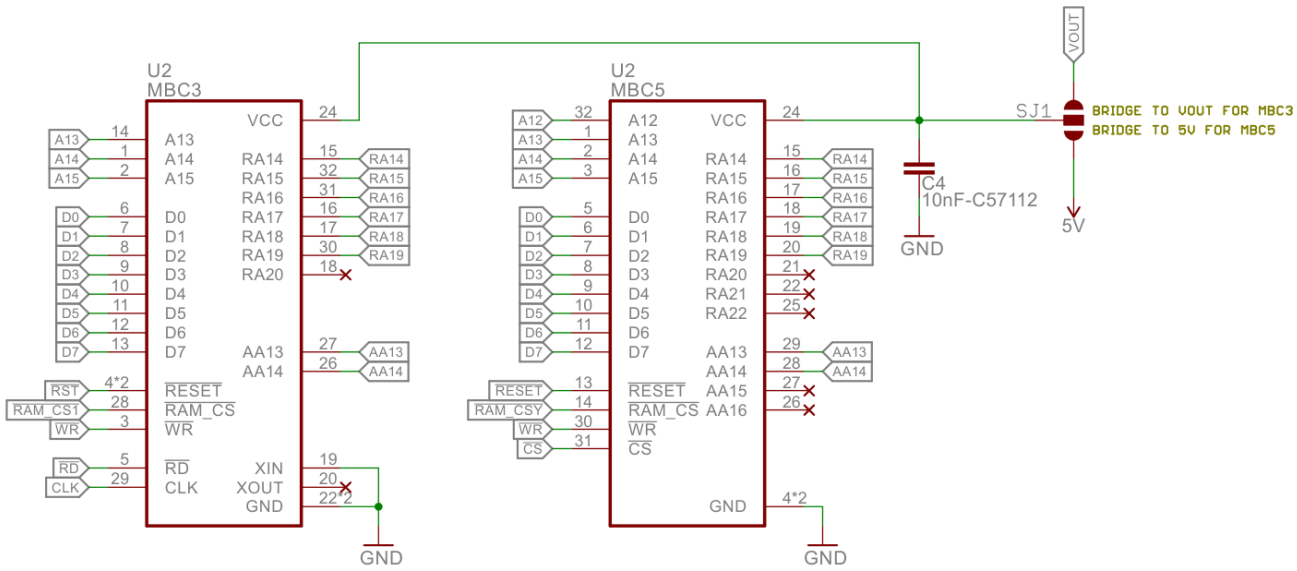
6. You can use a microfiber glasses-cleaning cloth or another soft material to gently press the sticker down and ensure it adheres evenly to the surface.

The giltesa's shop

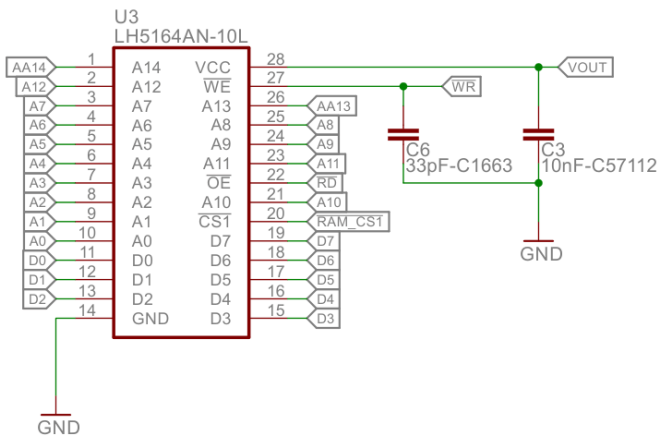
## SRAM-SUPERVISOR



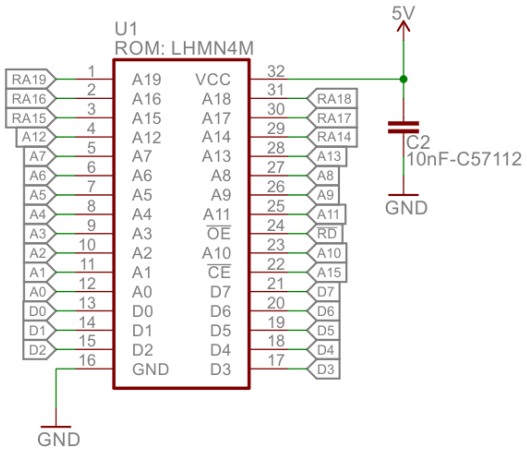

## MAPPER: MBC3 OR MBC5



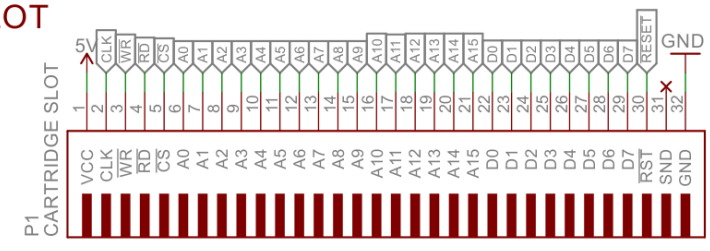

## SRAM




## ROM




## SLOT




TITLE:

# GB - POKEMON R.B.Y.

LICENCE:

https://giltesa.com

| | |
|---|---|
| FILE: Pokemon | REV: V1.1 |
| Date: 03/08/2026 22:55:54 | Sheet: 1/1 |

This PCB must have a thickness of 0.8 mm, ENIG surface finish, gold fingers, and 30° finger chamfer.